Homework 6: Concise Answers

1 (p. 1): Student seniorities

Assume four independent draws. Let n_i be the number of students from category i, with $\sum_i n_i = 4$. The unordered composition has probability

$$P(n_1, \ldots, n_5) = \frac{4!}{\prod_i n_i!} \prod_i q_i^{n_i}.$$

Laplace smoothing. If the given proportions are empirical frequencies from N students, add-one smoothing gives

$$q_i = \frac{Np_i + 1}{N + 5}, \qquad (p_1, p_2, p_3, p_4, p_5) = (.24, .22, .20, .18, .16).$$

The original gives no N, so the smoothed numerical probabilities depend on an unspecified sample size.

(i) All four are fourth-year students: $\boxed{q_4^4}$.

(ii) Two first-year and two second-year students: $\boxed{\binom{4}{2} q_1^2 q_2^2 = 6 q_1^2 q_2^2}$.

(iii) The most likely composition is $\boxed{(n_1, n_2, n_3, n_4, n_5) = (1, 1, 1, 1, 0)}$, with probability $\boxed{24 q_1 q_2 q_3 q_4}$. Indeed, if a category is repeated, there is an unused category. Moving one student from a repeated category i to an unused category j multiplies its probability by $n_i q_j / q_i > 1$, since $n_i \geq 2$ and $q_i / q_j \leq 1.5$. Thus the maximum uses four distinct categories, specifically the four largest.

(iv) The least likely composition is $\boxed{(0, 0, 0, 0, 4)}$, with probability $\boxed{q_5^4}$. Every product is at least q_5^4 and its multinomial coefficient is at least one.

Numerical example, assuming $N = 100$: $q = (25, 23, 21, 19, 17)/105$ gives, respectively,

$$\boxed{0.0010721541, \quad 0.0163203603, \quad 0.0452996437, \quad 0.0006871293.}$$

For comparison, without smoothing the four probabilities are

$$0.00104976, \quad 0.01672704, \quad 0.04561920, \quad 0.00065536.$$

2 (p. 2): Preparation given a correct answer

Let U mean unprepared and R mean a correct answer. By Bayes' rule,

$$P(U \mid R) = \frac{P(R \mid U)P(U)}{P(R \mid U)P(U) + P(R \mid \neg U)P(\neg U)} = \frac{(1/4)(1/4)}{(1/4)(1/4) + (0.9)(3/4)} = \boxed{\frac{5}{59} \approx 8.4746\%}.$$

3 (p. 3): Naive Bayes classification

Use the visible query $x = (\text{yes}, \text{yes}, \text{yes}, \text{yes})$ in the order (Give Birth, Can Fly, Live in Water, Have Legs). Let M and N denote mammals and non-mammals. There are 7 mammals and 13 non-mammals. The counts of each queried attribute value are:

Class	Birth=yes	Fly=yes	Water=yes	Legs=yes	Total
M	6	1	2	5	7
N	1	3	3	9	13

Using empirical probabilities and conditional independence,

$$s_M = P(M)P(x \mid M) = \frac{7}{20}\frac{6}{7}\frac{1}{7}\frac{2}{7}\frac{5}{7} = \frac{3}{343} \approx 0.00874636,$$
$$s_N = P(N)P(x \mid N) = \frac{13}{20}\frac{1}{13}\frac{3}{13}\frac{3}{13}\frac{9}{13} = \frac{81}{43940} \approx 0.00184342.$$

Therefore

$$\boxed{x \text{ is classified as a mammal}}, \qquad P(M \mid x) = \frac{s_M}{s_M + s_N} = \frac{43940}{53201} \approx 0.825924.$$

If add-one smoothing is also applied here, the binary attributes have two possible values, while Water has three (yes/no/sometimes). Smoothing the class priors as well gives

$$\widetilde{s}_M = \frac{8}{22}\frac{7}{9}\frac{2}{9}\frac{3}{10}\frac{6}{9} \approx 0.01257015,$$
$$\widetilde{s}_N = \frac{14}{22}\frac{2}{15}\frac{4}{15}\frac{4}{16}\frac{10}{15} \approx 0.00377104.$$

The smoothed classifier also predicts mammal, with posterior 0.769231.

4 (p. 4): Weather transition estimates

Use state order (S, C, R) for rows (current state) and columns (next state). The 17-day sequence gives 16 adjacent transitions:

From \ To	S	C	R	Row total
S	4	1	2	7
C	0	2	2	4
R	2	2	1	5
Total	6	5	5	16

Normalize each row to estimate $A_{ij} = P(W_{t+1} = j \mid W_t = i)$:

$$\boxed{A = \begin{pmatrix} 4/7 & 1/7 & 2/7 \\ 0 & 1/2 & 1/2 \\ 2/5 & 2/5 & 1/5 \end{pmatrix}}.$$

The last day's C has no observed successor and adds no transition.

5 (p. 5): Three consecutive sunny days

The given matrix, in state order (S, C, R), is

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}.$$

A three-day sunny sequence contains two transitions. For a window starting on day t,

$$\boxed{P(W_t = S, W_{t+1} = S, W_{t+2} = S) = P(W_t = S)(.8)^2}.$$

If day t is already known to be sunny, the conditional probability is 0.64. For a window starting on an arbitrary day in the stationary regime, solve $\pi A = \pi$, $\sum_i \pi_i = 1$:

$$20\pi_S = 38\pi_C + 75\pi_R, \qquad 40\pi_C = 15\pi_S + 5\pi_R, \qquad \pi = \frac{1}{929}(638, 245, 46).$$

Hence the stationary answer is

$$\boxed{\pi_S(.8)^2 = \frac{10208}{23225} \approx 0.43952637}.$$

An initial distribution is not specified. Outside stationarity the general formula above applies; if μ is the day-1 distribution, use $P(W_t = S) = (\mu A^{t-1})_S$.

6 (p. 6): Car-value Bayesian network

Let M, E, A, C denote Mileage, Engine, Air Conditioner, and Car Value. The graph gives

$$\boxed{P(M, E, A, C) = P(M)P(A)P(E \mid M)P(C \mid E, A)}.$$

The table represents 40 records. Summing the appropriate counts gives

$$P(M = \text{Hi}) = P(M = \text{Lo}) = \tfrac{1}{2}, \qquad P(A = \text{Working}) = \tfrac{25}{40} = \tfrac{5}{8}, \quad P(A = \text{Broken}) = \tfrac{15}{40} = \tfrac{3}{8}.$$

M	$P(E = \text{Good} \mid M)$	$P(E = \text{Bad} \mid M)$
Hi	$10/20 = 1/2$	$10/20 = 1/2$
Lo	$15/20 = 3/4$	$5/20 = 1/4$

E	A	$P(C = \text{Hi} \mid E, A)$	$P(C = \text{Lo} \mid E, A)$
Good	Working	$12/16 = 3/4$	$4/16 = 1/4$
Good	Broken	$6/9 = 2/3$	$3/9 = 1/3$
Bad	Working	$2/9$	$7/9$
Bad	Broken	$0/6 = 0$	$6/6 = 1$

For example, marginalizing M gives $P(C = \text{Hi} \mid E = \text{Good}, A = \text{Working}) = (3+9)/(3+4+9+0) = 3/4$. Since E and A are marginally independent in this network,

$$\begin{aligned} P(E = \text{Bad}, A = \text{Broken}) &= P(A = \text{Broken}) \sum_m P(m)P(E = \text{Bad} \mid m) \\ &= \frac{3}{8}\left(\frac{1}{2}\frac{1}{2} + \frac{1}{2}\frac{1}{4}\right) = \boxed{\frac{9}{64} = 0.140625}. \end{aligned}$$

This is the fitted-network probability; the raw empirical joint frequency is $(0+4+0+2)/40 = 0.15$.

7 (p. 7): Battery, fuel, gauge and starting

Let B, F, G, S denote Battery, Fuel, Gauge and Start. The graph has B, F as independent roots, and both are parents of G and S. Therefore

$$\boxed{P(B,F,G,S) = P(B)P(F)P(G \mid B,F)P(S \mid B,F)}.$$

In particular, G and S are conditionally independent given B, F. The supplied root probabilities are

$$P(B = \text{bad}) = .1, \quad P(B = \text{good}) = .9, \qquad P(F = \text{empty}) = .2, \quad P(F = \text{not empty}) = .8.$$

(a) Using the complementary start probability, $P(S = \text{yes} \mid B = \text{good}, F = \text{empty}) = 1 - .8 = .2$:

$$\begin{aligned} &P(B = \text{good}, F = \text{empty}, G = \text{empty}, S = \text{yes}) \\ &\quad = (.9)(.2)(.8)(.2) = \boxed{0.0288}. \end{aligned}$$

(b) Here $P(G = \text{not empty} \mid B = \text{bad}, F = \text{empty}) = 1 - .9 = .1$:

$$\begin{aligned} &P(B = \text{bad}, F = \text{empty}, G = \text{not empty}, S = \text{no}) \\ &\quad = (.1)(.2)(.1)(1) = \boxed{0.002}. \end{aligned}$$

(c) Marginalize the unknown fuel state. Since B and F are independent,

$$\begin{aligned} P(S = \text{yes} \mid B = \text{bad}) &= \sum_{f} P(S = \text{yes} \mid B = \text{bad}, f)P(f) \\ &= (1 - .9)(.8) + (1 - 1)(.2) = \boxed{0.08}. \end{aligned}$$

No gauge evidence is given, so its conditional probabilities sum to one and do not enter this marginal.